PRATHAM P PANSARE

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ABOUT ME

Aspiring software developer with hands-on experience building scalable, full-stack applications and AI-driven platforms. Proficient in modern web frameworks, relational databases, and machine learning integration, with a strong ability to translate complex technical requirements into user-centric solutions. Passionate about software architecture, clean code, and continuous learning.

EDUCATION

B.Tech in Information Technology
A.P. Shah Institute of Technology, Thane

Currently Studying | 4th Year

Higher Secondary Education
Jana Gana Mana College, Dombivli

Passing Out: March 2022 | Boards: 50% | MHCET: 83%

Secondary Education
Model English School, Dombivli West

Passing Out: March 2020 | Boards: 69%

TECHNICAL SKILLS

Languages:	Python, JavaScript, TypeScript, SQL, CSS, HTML
Frameworks:	React, Next.js, FastAPI, Flask, Tailwind CSS, Bootstrap
ML / AI:	TensorFlow, Google Colab
Databases:	PostgreSQL, MySQL, SQLite, Supabase
Cloud / DevOps:	AWS, Cloudflare R2, GitHub Actions, Vercel, HF Spaces
Tools:	Figma, Canva, GitHub

PROJECTS

ShetVaidya — AI Crop Disease Analyzer & Medicine Verifier

React, FastAPI, TensorFlow, PostgreSQL, Cloudflare R2

ShetVaidya ("field doctor" in Marathi) is an AI-powered platform designed to help farmers instantly diagnose crop diseases and verify medicine authenticity. The goal was to bridge the gap between rural agriculture and modern technology by providing real-time, location-aware diagnostic tools that reduce crop loss and prevent counterfeit medicine usage.

- Built a production-grade, location-aware crop disease analyzer and medicine verification platform to empower farmers with immediate diagnostic tools.
- Developed a TensorFlow MobileNetV2 inference microservice with prediction safety — dedicated non-leaf/garbage class rejection and confidence gating for out-of-scope images.
- Engineered a microservice architecture (frontend, backend, ML service) with FastAPI, JWT HttpOnly cookies, Google OAuth, and API rate limiting on auth and batch verification flows.
- Configured Cloudflare R2 object storage for secure image uploads (5MB limit, MIME validation) and integrated barcode scanning for medicine batch verification with non-public batch codes.
- Established CI/CD via GitHub Actions for containerized deployment across Vercel (frontend) and Hugging Face Spaces (backend + ML), with PostgreSQL for data persistence.

Khatawali Records — Credit/Debit Ledger App

React, Supabase, Bootstrap

Khatawali Records is a mobile-first ledger application built for small business owners and individuals who struggle with tracking credit and debit transactions. The app simplifies financial record-keeping by enabling category-separated tracking, quick contact imports, and secure per-user data isolation — all packaged as a native Android app.

- Engineered a mobile-first credit/debit ledger application for small business owners and individuals, with category-separated tracking for the same person across independent ledgers.
- Packaged natively for Android with deep-link OAuth callbacks, in-app contact importer, and biometric fingerprint lock configurable via app settings.
- Designed a PostgreSQL schema on Supabase with Row Level Security (RLS) policies and Supabase Auth (Google OAuth) to guarantee per-user data isolation and privacy.
- Built a dashboard with advanced search (name, phone, description, category), date-range and year/month filtering, and quick actions for call, WhatsApp, receive, give, and transfer.
- Integrated per-person PDF and Excel export using autotable and PapaParse, along with profile/business details, bank holidays, recycle bin, and developer support.

Scholar Nexus — AI Research Discovery Platform

Next.js, TypeScript, Prisma, D3.js

Scholar Nexus addresses the challenge of fragmented academic research by aggregating papers from 9 major databases into a single intelligent platform. It helps researchers discover relevant literature faster through AI-driven query expansion, smart deduplication, and visual citation networks — turning hours of manual searching into minutes of guided exploration.

- Architected an AI-powered research paper discovery platform that simultaneously queries 9 academic databases (Semantic Scholar, arXiv, Crossref, PubMed, OpenAlex, and more) in parallel.
- Implemented a 10-step search pipeline: query understanding, AI-driven expansion, parallel fanout, normalization, deduplication (DOI + title-hash + fuzzy), ranking, and AI insights.
- Developed composite ranking scoring papers across citation impact, recency, source authority, and open-access availability, with AI-generated per-paper insights (TL;DR, methodology, limitations).
- Built interactive D3.js citation network graphs, PDF full-text Q&A; chatbot, evidence synthesis across papers, color-coded reading collections, daily/weekly search alerts, and author profiles.
- Engineered 18 API route handlers with rate limiting (token bucket), SSRF protection, Zod input validation, and Prisma parameterized queries; deployed on Vercel with GitHub Actions.

Air Pollution Tracker — Environmental Intelligence

React, Tailwind CSS, Supabase, React Leaflet, Recharts

This project tackles the lack of accessible, real-time air quality data for citizens and government bodies alike. By visualizing live AQI metrics for 200+ cities with role-based portals, it empowers citizens to make health-informed decisions while giving officials the analytical tools needed for environmental policy and incident response.

- Created a data-driven React application visualizing live and historical air-quality metrics for 200+ cities worldwide, with OpenAQ v3 integration and EPA breakpoint AQI computation.
- Implemented differentiated role-based portals: citizen experience (dashboard, AQI tracking, map view) and government portal (heatmaps, incident desk, pollutant intelligence, reporting panels).
- Built interactive map views with React Leaflet clustered markers and geolocation (browser API with IP-based fallback via ipwho.is), plus Phase 5 analytics for single-city deep dives and forecasting.
- Designed a robust data layer using React Query v5 with stale-while-revalidate caching, 30-minute localStorage cache for OpenAQ snapshots, and virtualized list rendering via react-window.
- Integrated Supabase Auth with domain-guardrailed registration (citizen: @gmail/@outlook, government: @gov), email OTP verification, offline-friendly preferences, and audit logging for government.